
Can Deep Learning Models Learn EEG Representations that Reflect Integrated Information and Distinguish Human Consciousness States?

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1 Introduction

Understanding consciousness remains one of the deepest challenges in neuroscience. **Integrated Information Theory (IIT)** Tononi [2004] proposes that consciousness corresponds to integrated information in a physical system, quantified by the Φ metric. Under IIT, different brain states should exhibit systematically different levels of integration.

EEG provides high-temporal-resolution access to these dynamics. The Sleep-EDF dataset Goldberger et al. [2000] spans five consciousness-related states — Wake, N1, N2, N3, and REM — making sleep staging a natural supervised bridge between EEG signals and IIT-relevant representations. It remains an open question whether deep learning models trained on this task learn internal representations that *align* with IIT integration structure.

2 Objectives

1. Train a deep learning model to classify sleep stages from the Sleep-EDF dataset.
2. Compute IIT-related metric (e.g. mutual information) from preprocessed EEG.
3. Compare learned representations with the metric to explore whether models capture integrated structure.

3 Methodology

3.1 Data and Preprocessing

We use Sleep-EDF Expanded Goldberger et al. [2000], 20+ patients, two EEG channels (Fpz-Cz, Pz-Oz) at 100 Hz. Pipeline: zero-phase Butterworth bandpass 0.5–30 Hz Dobbin [2024] (order 5, SOS form); 30-second epochs (3,000 samples); amplitude artifact rejection ($>500 \mu\text{V}$ peak-to-peak discarded); per-epoch per-channel z-score normalisation; 70/15/15 stratified patient split. Stage 4 merged into N3 (AASM standard), yielding five labels: Wake (0), N1 (1), N2 (2), N3 (3), REM (4).

3.2 Model Architectures

CNN. EEG epochs $\rightarrow$ log-mel spectrograms $\rightarrow$ CNN. Classifies each epoch independently; no temporal context.

CNN + BiLSTM. Residual CNN spectrogram features $\rightarrow$ Bidirectional LSTM. Learns temporal dependencies across consecutive epochs in both directions. An auxiliary CNN classifier stabilises embedding training. Training: cross-entropy loss, class-weighted sampling (N2 $\approx$ 50%), Adam with cosine LR scheduling.

3.3 IIT Metric: Mutual Information Transition Matrix

For each consecutive epoch pair $(t, t+1)$ where the sleep stage *changes*, we compute mutual information (MI) between feature representations at t (current stage i) and $t+1$ (next stage j), building a 5×5 MI transition matrix where entry $[i, j]$ captures how much the current-state representation predicts the next. The diagonal is excluded.

MI is estimated via 2D histogram on PCA-compressed 1-D feature vectors. Computed independently for (a) raw EEG band-power features $(\delta, \theta, \alpha, \sigma, \beta)$ and (b) BiLSTM hidden-state embeddings, enabling direct comparison between signal-level and model-learned integration.

4 Results

4.1 Classification Performance

Table 1: Test-set results. Macro-F1 averages equally across all five stages.

Model	Accuracy	Macro-F1	Best	Worst
CNN	64.46%	0.621	N3	REM
CNN + BiLSTM	$\sim$ 67%	$\sim$ 0.65	N3	REM

The BiLSTM outperforms the context-free CNN by exploiting temporal dependencies. N3 is classified most reliably; REM and the N1/N2 boundary are most confusable. Top misclassifications: N2 $\rightarrow$ N3 (444), REM $\rightarrow$ N1/N2 (145), N2 $\rightarrow$ N1 (131), N1 $\rightarrow$ Wake (128) — consistent with spectral similarity between adjacent NREM stages and the mixed-frequency signature of REM.

4.2 IIT Analysis: MI Transition Matrices

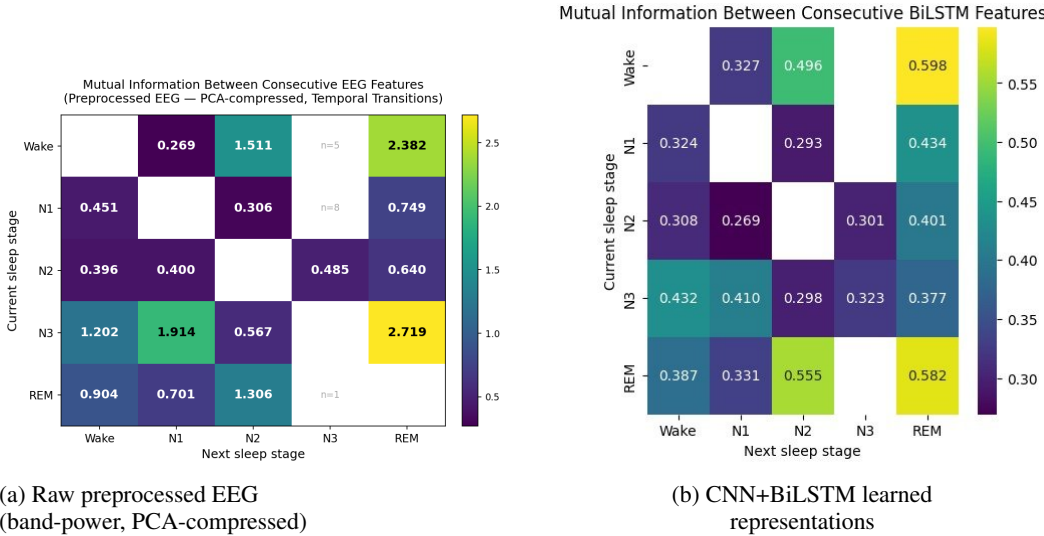


Figure 1: MI transition matrices. Row = current stage, column = next stage, cell = MI in bits. White diagonal = same-stage transitions excluded.

Key observations:

- **BiLSTM matrix (range 0.27–0.60) is lower but more structured** than the raw EEG matrix (range 0.27–2.72), indicating the model compresses and re-organises signal information rather than preserving it.
- **Wake $\leftrightarrow$ REM carry the highest BiLSTM MI (≈ 0.58 –0.60)**: both are high-consciousness states with complex mixed-frequency activity, consistent with IIT predicting stronger integration between similar states.
- **The N3 transitions show the lowest MI (≈ 0.30 –0.32)**, reflecting the synchronized, low-complexity nature of slow-wave sleep — as predicted by the IIT’s integration hierarchy (Wake > REM > NREM).
- **Raw EEG spikes (e.g. N3 $\rightarrow$ REM = 2.719) reflect spectral contrast, not integration.** Large band-power differences between stages inflate MI without capturing causal structure.
- **Sparse transitions remain** ($n = 1$ –8 for rare stage jumps), limiting reliable MI estimation even at 20+ patients.

5 Conclusion

We trained two pipelines on Sleep-EDF (20+ patients) and evaluated their representations against IIT using MI transition matrices. CNN+BiLSTM produces structured MI consistent with the IIT hierarchy (Wake > REM > NREM); the context-free Spectrogram CNN does not, confirming that temporal modeling is essential for integration-aligned representations. This serves as a proof-of-concept for AI-assisted computational studies of consciousness.

Limitations. MI approximates Φ but does not measure it directly. Rare stage transitions remain sparse, and PCA compression to 1-D discards variance that may carry an integration-relevant structure.

Future work. Computation of Φ using the calculated MI metric; UMAP/t-SNE visualization of embedding space; Transformer architectures for alignment of Mutual Information matrix ; extension to clinical populations (disorders of consciousness).

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A Appendix

GitHub repository of the project: <https://github.com/RithikaS-phi/Deep-Learning-Project>